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Measuring forest damage of ungulates: what should be considered

Friedrich Reimoser^{a,*}, Helen Armstrong^b, Rudi Suchant^c

^a*Institute of Wildlife Ecology, Veterinary University, Savoyenstrasse 1, A-1160 Vienna, Austria*

^b*Scottish Natural Heritage, 2, Anderson Pl., Edinburgh EH6 5NP, UK*

^c*Forstliche Versuchs- und Forschungsanstalt Baden-Württemberg, Wonnhaldestrasse 4, D-79100 Freiburg, Germany*

Abstract

Principles and criteria for the objective assessment of ‘damage’ are presented as a basis for objective discussion and to promote targeted research. The topic will be considered from three different viewpoints – theoretical background, silviculture, and nature conservation. Examples of methods to assess ‘damage’ are given, as are the criteria and indicators for target values for growing stock.

The objective existence of damage must be established by comparing the current status against a target. Only if the present status no longer permits achievement of the desired target can damage be claimed. The same is true for verification of ‘benefit’ resulting from an impact leading to a favorable status. Establishing the degree of browsing damage on forest regeneration allows forecasts of damage to be made, but the actual damage can only be determined at a future time, for example when the timber is harvested or when the function of the forest is seen to have been compromised. The earlier the forecast is made the more uncertain it is, because the compensatory responses of the forest can only be forecast to a limited extent. Silvicultural targets depend on the requirements of both, the forest owner and the public. Main indicators for assessing ungulate damage on forest regeneration are total tree density, species composition, and height structure. From the viewpoint of nature conservation the determination of target values is particularly difficult. The measurement of damage involves choosing the most relevant indicators of favorable condition of woodland habitats and assessing these in the field. With adequate resources this can be done quantitatively using a wide range of standard field measurement techniques. Alternatively, where resources are limited, it can be done qualitatively by scoring different areas for a range of indicators. As yet, such qualitative methods are in their infancy. Specific research is now being undertaken in some European countries. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

It is recognized that ungulate herbivores can have a profound effect on the vegetation and soil of forests and woodlands (e.g. see Eiberle and Nigg, 1983;

Putman, 1986, 1996; Ammer, 1996). Damage by twig browsing and bark peeling is an increasing problem in many European countries and elsewhere (e.g. see Mitchell et al., 1977; Mayer and Ott, 1991; Donaubauer, 1994). However, there are considerable difficulties in objectively assessing the damage done by ungulates (Schwarzenbach, 1982; Gill, 1992; Reimoser, 1986; Reimoser et al., 1997). Hasty and false

*Corresponding author. Tel.: +43-1-489-0915-55; fax: +43-1-489-0915-59; e-mail: friedrich.reimoser@vu-wien.ac.at

inferences about damage frequently result in conflicts between foresters, landowners, hunters, nature conservationists, federal authorities, and even tourists.

A set of principles and criteria for the objective assessment of ‘damage’ is presented as a basis for objective discussion and to promote targeted research. The topic will be considered from three different viewpoints – theoretical background, silviculture, and nature conservation. Examples of methods to assess ‘damage’ as well as of criteria and indicators for target growing stock are given.

2. Theoretical and basic aspects

2.1. Is it damage, merely an impact, or even a benefit?

An objective and realistic assessment of damage is difficult, particularly as regards browsing on natural regeneration. Different estimates have given different results (Reimoser, 1986). Why is this? The problem often relates to an absence of an operational target for growing-stock. Damage (in the sense of ‘a problem

caused by an unwanted condition’) is an anthropocentric concept used in relation to one or more specific species. To meaningfully ascertain damage in an ecological system, however, requires that a concrete aim – a desired condition – be defined and compared with the current condition in order to determine whether ‘damage’ or merely an impact or disturbance by game has in fact occurred. This, in turn, requires that operational limits and critical loads be specified. For example, the browsing of 1000 trees/ha may be a damage when we have only 2000 trees/ha. It is no damage when we have 10 000 trees/ha and we need for the future only 1500/ha. From a species-neutral perspective, the term damage has no meaning. A problem needs an owner – it is the problem owner who defines damage. The same is true for verification of ‘benefit’ resulting from an impact leading to a favourable status, e.g. if, through selective browsing of strong rival species, weaker species are enabled to achieve target levels.

Assessment of damage or benefit must be made following a systematic approach of the form shown in Fig. 1. The upper level reflects the ascertainment of the current status and the recognition of an ‘impact’. It

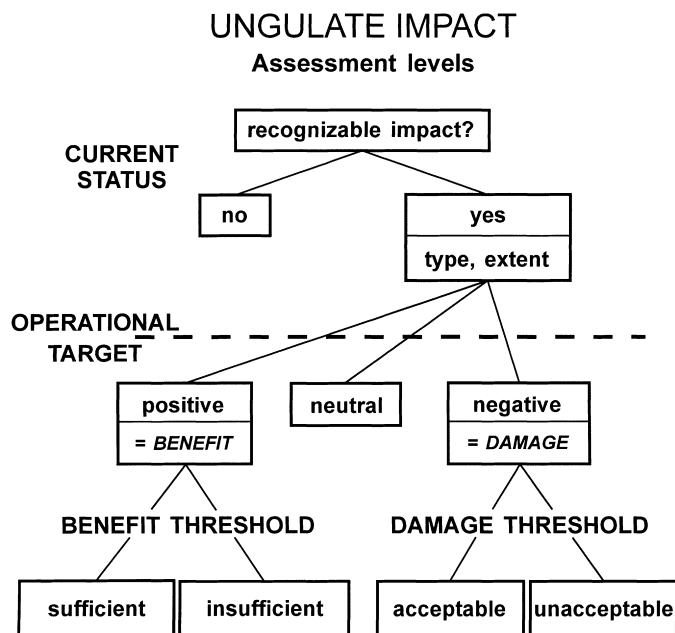


Fig. 1. Levels for analyzing impact, benefit, and damage.

is the level noted without evaluation in terms of a specific viewpoint. However, an analysis of current states and impacts alone cannot lead to valid estimates of ‘damage’. It is only when an operational target has been explicitly stated that the actual status can be related to the probability of achieving that target stock, i.e. whether impact is positive (beneficial), negative (damage) or neutral. Socio-economic aspects and subjective valuations play an important role in such assessments. Further differentiation of the valuation system (lower level, Fig. 1) makes possible the definition of a damage threshold above which the level of game damage is no longer acceptable, or a benefit threshold favourable for achievement of the target for a particular growing stock.

The various ways in which ungulates can impact on forest vegetation comprise:

- (i) trampling (includes pawing, scraping, burrowing and rooting);
- (ii) browsing (includes ‘unseen’ browsing, i.e. feeding on seeds and seedlings, and ‘visible’ browsing, e.g. tree twig browsing);
- (iii) fraying; and
- (iv) peeling (e.g. of bark or surface roots).

Damage to forests, as opposed to impacts, embraces concepts such as loss or reduction of increment, economic value, ecological stability, diversity, sustainability, and the value of a forest for avalanche or rock-fall protection, etc. Benefits include aspects such as increase in diversity, stability, gain in economic value (e.g. resulting from selective browsing of unwanted plant species).

Target values for regeneration (e.g. lowest acceptable tree stem count, tree species distribution, period for reaching 1.5 m height – top twig beyond browsing range) may be determined in terms of forest-communities and forest functions for each type of stand (Reimoser and Suchant, 1992; Erhart, 1994; Schulze, 1997; Reimoser et al., 1997). In the establishment of a damage threshold (e.g. browsing tolerance limits), it is essential to distinguish whether regeneration targets are set, for example, in terms of forest industry requirements (e.g. optimization of forest income), or in social terms (e.g. sustained protective forest function or landscape design). On account of differing viewpoints, targets and thresholds, the estimate of what constitutes damage or benefit can be markedly

different, even with identical levels of ungulate impact.

The effect of ungulates (positive, negative or neutral) can best be judged by comparing the lowest acceptable regeneration with, and without, ungulate impact, e.g. by comparing growth inside an ungulate-proof fenced area with that outside, thereby obtaining two related current-status values. Both these values need to be compared with the operational target. It is not enough merely to compare the two current status values, since the regeneration status in the protected area is neither natural (i.e. there are no ungulates!) nor need it be the desired status for the forest (Reimoser and Suchant, 1992).

2.2. *Is correct forecasting of damage possible?*

The determination of browsing damage on forest regeneration at a given time forms the basis for a forecast, because the real damage can only be determined at a future time, for example when the timber is harvested or when the function of the forest is seen to have been compromised. The earlier the forecast, the more uncertain it is, because the compensatory and regenerating reactions of the forest can only be forecast to a limited extent. Cause and effect, i.e. browsing and the resulting damage, can be decades apart. This makes it difficult to estimate damage and benefit at the time of browsing. This is very different to the situation in farming, where the damage normally occurs within one year, making it easier to estimate. In order to obtain a good approximation of the final status of the growing stock as a result of browsing impact, indicators have to be specified for young forest stands at the time when top-twig browsing is no longer possible.

2.3. *Existing concepts*

It is essential to differentiate between methods (and studies) that focus on single trees or those that focus on regeneration groups. The former focuses on numbers of trees, etc. while the latter considers groups of young trees (regeneration nuclei) in terms of numbers, distribution and development of different groups. Such group-oriented methods have, above all, been developed for mountain forests with regeneration patterns based on an irregular distribution of small canopy gaps (Näscher, 1979; Ott, 1998). This review, however,

primarily focuses on the individual-oriented methods (single trees).

Methodological approaches extend from focus on young tree mortality (e.g. percent browsed, see Eiberle and Nigg, 1987; Odermatt, 1996) to the conceptual opposite of disregarding mortality counts and concentrating on the number undamaged (Reimoser, 1986; Roth, 1995; Schulze, 1997; Reimoser et al., 1997).

Cost-oriented tables for assessing browsing and other game damage prepared for forest owners in order to claim compensation from hunters (e.g. Polanschütz, 1995; Speidel, 1980; Moog and Niebler, 1995; Gundermann and Suda, 1994) will not be considered here, because they mostly do not consider damage in relation to an operational target.

2.4. Damage by ungulates – steps of an objective diagnosis

Despite the uncertainties described above (Section 2.2), it is possible to objectify damage assessment to some extent. Objectification in this context is taken to mean all the methods that restrict subjectivity in assessment in the interests of avoiding misunderstandings and conflicts between interested parties. In this connection, use of a fenced control area for comparison is useful though not sufficient (Reimoser and Reimoser, 1997).

When does a disturbance of vegetation by ungulates become damage? Not every twig browsed represents damage to a tree; not every tree damaged represents damage to a stand. To ascertain objectively the existence of ungulate damage, the following steps should be specified (Reimoser and Gossow, 1996):

2.4.1. Is there a need for stand regeneration?

For example, after a pole-stage stand has regenerated for an appropriate period after thinning, the regeneration dies off as the canopy of the stand grows dense again. Browsing cannot be counted by the forester as ‘damage’ when any regeneration would die even without browsing. In other words, the browsing impact is now included within ‘compensatory mortality’ and, therefore, does not affect the further development of the stand. Some decades later, when the stand needs to regenerate again, the same browsing impact may truly be ‘damage’. The concept of compensatory mortality is usually discussed in connection

with animal populations, but it is at least as important for plant populations.

2.4.2. The operational regeneration target

A target set at the planning stage might, for example, be 3000 young trees per hectare undamaged with a distribution of at least 20% spruce, 10% fir, 20% beech, the remainder being of the same or other tree species. If, after disturbance by game, there are still enough undamaged trees to fulfil the target, the browsing will not have reached the level of ‘damage’. To reiterate, damage to trees does not automatically qualify as damage to a stand. Thus, when calculating ungulate damage, one has at first to focus on the number of undamaged trees. The mortality of trees is less important than the number that survive.

2.4.3. The response of trees to browsing

The effect of browsing depends on the tree species, site conditions, time and intensity of browsing, etc. (Canham et al., 1994). Again, not every twig browsed qualifies as damage to the tree.

2.4.4. Determination of the damaging species

Is the impact truly being inflicted by ungulate animals or by other species (e.g. hare, mice, humans)? Only ungulate animals can effect ‘ungulate damage’, but there are many causes of damage to trees, often with very similar symptoms. If game damage has to be assessed, the impact of game ungulates must be differentiated from the impact of domestic ungulates (e.g. cattle, sheep).

This objective diagnosis has been incorporated into methods for browsing-damage assessment in Vorarlberg, Austria (Reimoser and Suchant, 1992; Reimoser et al., 1997), in Baden-Württemberg, Germany (Roth, 1995), and in Hessen, Germany (Schulze, 1997).

2.5. Benefit deriving from ungulates

Possible beneficial impacts of ungulates range from the treading-in of seeds into the ground and their dispersal, through selective browsing of unwanted competing species (e.g. blackberry competing against tree-species; Reimoser, 1986) to improving regeneration conditions as a result of their droppings and redistribution of nutrients. However, only little research data concerning the positive effects of ungulates in the forest ecosystem exist (e.g. Putman, 1986;

Reimoser, 1986; Wolf, 1988) and, in contrast to the situation in respect of the negative impacts, positive ones have rarely been sought. Benefits to the forest from ungulates have also hardly been recognized in forestry practice – indeed this has been considered to be impossible.

Recent studies have shown evermore clearly that, apart from the key interactions competition and predation, herbivory and, in particular, mutualism also have a marked effect on natural communities. The research into plant-animal interactions provides new insights into ecology and opens possibilities undreamt of a few years ago (Howe and Westley, 1993). It is in this sense that the forest-ungulate interactions must be seen and further researched. It is also of vital and practical importance to know under what conditions (from an anthropocentric perspective) can ungulates have positive impacts, and how these can be optimally utilized.

2.6. Ungulate impact on forest regeneration – forest management impact on damage by ungulates

The impacts of ungulates on tree species have variously been found to result in:

- (i) a decrease in diversity and/or abundance;
- (ii) an increase in diversity and/or abundance;
- (iii) changes in structure without change in diversity or abundance; or
- (iv) no ascertainable influence (Reimoser and Gossow, 1996).

Which of these reactions was realized at a certain place and time basically depended on (i) the type of 'disturbance' (e.g. the nature, intensity and duration of the impact of ungulates on soil and plants), and (ii) the 'reaction' of the respective system (e.g. soil and plants). However, the type of reaction depended on the initial situation at the time of the disturbance (soil status, germination conditions, vegetation density, species composition, browsing attraction and browsing tolerance of plants, competition between plant species, existing seed trees, light and growth conditions, direction of development, etc.). In turn, the initial situation depended strongly on the type of forest management.

Thus, the effect of ungulates – even when they are a constant 'disturbing factor' (e.g. same deer density) – can lead to totally different results. For example, an

increase of diversity and abundance might turn into a decrease or vice versa. Thus, the direction of development of an ecosystem, its dynamics and possible feedbacks, can be markedly modified: human impacts (e.g. silvicultural measures, pollution) often provide the impetus that changes the effect of ungulates on the ecosystem (Reimoser, 1986; Ellenberg, 1988).

2.7. Benefit/damage relationship

The relationship between benefit from and damage by ungulate game on natural forest regeneration was studied in two Austrian regions having roe deer (*Capreolus capreolus*), red deer (*Cervus elaphus*) and chamois (*Rupicapra rupicapra*) (Reimoser and Reimoser, 1997). Both these regions were dominated by montane mixed forest (particularly spruce, fir, beech and mountain maple). The survey method (fenced control vs. unfenced area, Reimoser and Suchant, 1992) and the analytical method (current status/target stock comparison, Erhart, 1994; Reimoser et al., 1997) were the same for both the regions. Indicators, target values, an tolerance limits are shown in Table 1.

About 800 test areas were studied over a period of six years. In both the studies, there were areas on which the indicators exclusively showed game damage, and others with exclusive benefit, though the ratio of damage to benefit varied from about 4 : 1 to 14 : 1.

Whether ungulate game impact – based on the test criteria (Table 1) – results in net benefit or net damage reflects also, in addition to targets and ungulate population, the predisposition for damage or benefit of the forest regeneration itself. This predisposition is markedly affected by forestry management practices (Reimoser and Gossow, 1996). Both, forestry and ungulate-game management must be deliberately targeted to obtain the most positive interaction between game and habitat structure.

3. Silviculture and game damage

Depending upon the silvicultural goals that are pursued, impacts of game on the forest may be considered damage or not. But how are silvicultural goals defined and set?

Table 1

Indicators with operational targets and intolerance limits used in the monitoring system of the Austrian province of Vorarlberg until 1991 (Reimoser et al., 1997). The limits for height increment loss (*H*) for slow growing regeneration (maximum annual growth <10 cm) was loss of two, and for rapidly growing three height classes (out of eight). Multiple top twig browsing was used as an auxiliary (i.e. early warning) indicator for height increment loss. Shrub growth was also taken into account in some forest communities in which shrubs play a vital role with respect to preserving the productive power of the soil. Damage is taken to exist when any one indicator (a target value or a tolerance limit) is exceeded due to ungulate game (comparison between fenced control and unfenced areas). Benefit exists when at least one indicator is achieved due to ungulate game, e.g. one key tree species achieves a height increment gain of more than two height classes (or three for rapidly growing species), e.g. by browsing of rival tree species

	Indicator	Target value ^a
<i>N</i>	regeneration density	2000–5000 trees/ha
<i>C</i>	composition type (deciduous, coniferous, mixed)	minimum 10–50% of regeneration density (<i>N</i>) are deciduous and/or coniferous
<i>K</i>	key tree species	minimum 10–20% of regeneration density (<i>N</i>)
<i>T</i>	tree-species number	minimum 1–4 species
<i>I</i>	shrub-volume index ^b	minimum 600 m ³ /ha equiv.
Intolerance limit		
<i>H</i>	height increment ^c	difference 2–3 height classes ^d
<i>S</i>	shrub-species number ^b	40% difference
<i>B</i>	multiple top-twig browsing ^e	30% of trees

^a Depending on anticipated natural forest community and forest function.

^b Some forest communities.

^c Highest trees of regeneration.

^d Depending on maximum top-twig length.

^e Multiple top-twig browsing on highest trees during a period of three years.

3.1. What is silvicultural game damage?

In determining game damage, it is necessary to differentiate between the effects of bark stripping (peeling) and the effects of browsing and fraying. Peeling occurs when game bites and tears accessible parts of the bark. This often results in large wounds which, in turn, may lead to: wound rot; grading losses; increased harvesting costs; increased manipulation costs; degradation of the diameter structure of the standing stock within a stand; additional management measures; increased danger of blowdown and snow breakage; shortening of rotation length; artificial instead of natural regeneration; and failure to achieve the management goal.

During browsing, the terminal shoots and the side shoots as well as the buds of young plants are eaten, while fraying (rubbing) injures the stems and branches of young trees. Damage from browsing and fraying may lead to: decrease in growth rate, deformation of trees, increased replanting, additional management measures, requirement for artificial regeneration, loss of mixture due to selective browsing, failure to reach the management goal.

3.2. Silvicultural goal setting

Silvicultural goals are determined by the needs and demands that the forest owner and the general public place on the particular forest. The requirements of the forest owner could be wood production based on quantity and quality, reduction of risk of damage, sustainable forest utilization, preservation of the variety of forest products, and so forth. The general public also expects to fulfil forest management requirements. Correspondingly, goal setting differs from forest to forest, from region to region and from country to country. These differences also cause differences in forest management: type of regeneration, choice of tree species, rotation length, desired forest structure, extent of risk, sustainability of forest functions, intensity of management, and ecological harmony. The extent of game damage essentially depends upon the silvicultural goal setting and on the type of management derived from the goals.

When setting silvicultural goals, it is necessary to differentiate between planting and natural regeneration. Measurable targets that allow an easy and sufficient determination of damage are set when using

artificial regeneration. In this case, the silvicultural target is reflected directly in the planting method. The forest owner selects the tree species to plant, the number of plants and their spacing. Game damage is recognized if the current target is not attained.

Silvicultural goals for natural regeneration have been very vague in the past, only verbally described and without measurable factors. The regeneration goal is often described in terms of desired tree species relationships. Measurable parameters, such as the number of plants, tree species relationship according to height layers or height structure, are not given (e.g. Weidenbach, 1990). An additional problem is that the verbally stated goal for the tree species relationship cannot always be achieved even if no game impact occurs. This makes evaluation of game damage in natural regeneration areas very difficult, and evaluation requires an objective procedure that integrates the various characteristics of natural regeneration (e.g. Roth, 1995).

3.3. *Silvicultural targets, game damage, site and silviculture*

The framework for setting of silvicultural targets is determined by site differences. Site factors have a decisive importance and influence tree species, growth and regeneration potential. The site primarily influences whether regeneration occurs, the number and the vigorousness of tree species regenerating, and the length of exposure to game impact. The supply of alternative grazing for game is also influenced by the site. How much game requires trees as a plant food source, can be influenced by the alternative grazing supply (Reimoser, 1986).

Site differences, however, can be over-ridden by silvicultural management. The decisive factor for regeneration development is the light and climate conditions associated with it. Light decisively determines the height development (Ammer, 1996; Roth, 1995). With minimal available light, regeneration for some trees is difficult. The height development also determines the period during which the plants are endangered by browsing (risk period). Thus, the silvicultural system determines, based on light supply, the height development and the associated risk period which, in turn, influences the degree of damage. Furthermore, the alternative grazing supply is also

dependent on the intensity of the available light and silvicultural management.

Further differences in respect of game damage evaluation must also be considered; for example, the historical development of the forest and the difference between the current condition and the desired silvicultural target. All of these factors produce a complex interaction that should be considered when making a silvicultural evaluation of game damage.

An example of evaluation of silvicultural damage to natural regeneration by defining three parameters (plant density, tree species mixture and height structure) for fenced (control) and unfenced areas was given by Roth (1995). His step-wise evaluation procedure answers the following questions:

- (i) Does browsing affect the development of forest regeneration?
- (ii) Can the silvicultural targets be achieved without game impact?
- (iii) Are the silvicultural targets being reached even with game impact?

Using this procedure, silvicultural damage will only be assumed proven if any one minimum value is satisfied for the fenced area but not for the unfenced area. In this work, the browsing percentage was shown to be unsuitable for determining silvicultural damage. For example, there were areas with an average of >50% browsing impact which, nevertheless, did not qualify as silvicultural damage, while other areas with <10% browsing impact were recognized as representing damage.

4. *The definition and measurement of damage to nature conservation value*

4.1. *What is nature conservation value?*

Before we can define damage to the nature conservation value of a woodland, it is necessary to define what constitutes a favourable condition. Damage is then any change which induces a condition that can no longer be described as favourable.

A favourable condition of a habitat has been defined as occurring when “the specific structure and functions which are necessary for its long-term maintenance exist and are likely to continue for the

foreseeable future and the conservation status of its typical species is also favourable” (UK Monitoring Network and Natura 2000 Coordinators Group, 1997, unpublished). If it is assumed that a decision has been made that a particular woodland type is to be conserved, how far can one go towards defining generic indicators of the condition of the habitat? To what extent are such indicators a function of the level of grazing and browsing by ungulates? Can it be assumed that, if the plant component can be ‘correctly’ defined, that this will also ‘get the animal component right’?

4.2. Indicators of favourable condition of woodland habitats

Tucker et al. (1997) have suggested indicators of favourable condition for woodlands found in the UK that are listed as priority habitats for conservation in Annexure 1 of the European Community’s Habitats and Species Directive. Table 2 lists the indicators for each woodland type which relate most directly to grazing and browsing pressures. These indicators relate to rates of tree regeneration either from seed

or by resprouting, depending on the type of woodland. Different amounts of regeneration are desirable for different types of woodland, with the least regeneration needed in wood pasture systems. The location of regeneration with respect to the density of the canopy depends on the shade tolerance of seedlings and saplings. The authors do not specify any indicators specifically associated with the ground- or shrub-layer vegetation. However, in some cases, there is a correlation between indicators of ground and/or shrub-layer condition and amount of tree regeneration, and it may be easier to use such indicators than to judge or measure the extent of tree regeneration itself. Conversely, it may be that, if sufficient tree regeneration is being achieved then the majority of the plant species found in the ground and shrub layers associated with the habitat will also be in a favourable condition, and that may also be true for the animal species.

Mitchell and Kirby (1990), Reid (1996), and Reynolds et al. (1997) (personal communication) have produced lists of generic indicators of very high, high, moderate, low and no grazing pressure in woodlands (Table 3). These indicators are based largely on easily

Table 2

Generic indicators of favourable condition of woodlands which relate to grazing and browsing levels (from Tucker et al., 1997, unpublished). In addition, an indicator given for all woodland types is that populations of characteristic plant and animal species, which may be specific to each site, should be viable

Woodland type	Sub-type	Indicator influenced by grazing and browsing levels
Tilio-acerion ravine forests	natural	gap-phase regeneration should be taking place in all gaps more than five years old and as advance regeneration below mature, unbroken stands
	managed high forest	minimal tree damage from deer regeneration should be wholly by natural regeneration or regrowth from stump
	coppice	gap-phase regeneration as for natural sub-type grazing levels controlled if necessary to permit coppice regrowth to form closed canopy.
	wood pasture	grazing and browsing should be reduced to allow regeneration for one period of at least 15 years in every 100 years over 50–75% of the ground
Caledonian forest	natural and managed	grazing and browsing limited to permit established saplings in gaps to develop into trees
Old oak woods with <i>Ilex</i> and <i>Blechnum</i> in the British Isles	Near-natural and managed high forest	grazing and browsing allows regeneration in shade and in gaps
	coppice	grazing and browsing levels permit regrowth to form closed canopy
	wood pasture	grazing and browsing reduced for one period of at least 15 years in every 100 years over 50–75% of ground
Residual alluvial forest	floodplain woodland	grazing and browsing allows regeneration on open ground
	swamp forest	grazing and browsing allows regeneration in gaps and on open ground
<i>Taxus baccata</i> woodland		none set
Bog woodland		none set

Table 3

Indicators of different grazing or browsing pressures in woodland (taken from Mitchell and Kirby, 1990, and Reid, 1996, with additions from Reynolds et al., 1997, personal communication)

Very heavy – No shrub layer; obvious browse line on mature trees; ground vegetation <3 cm tall with grasses, mosses or bracken (*Pteridium aquilinum*) predominating, and trampling down of ground flora; extensive patches of bare soil; surviving herb species usually dominated by unpalatable species, such as wood sorrel (*Oxalis acetosella*) and bluebell (*Hyacinthoides non-scripta*); suppression of growth, and killing, of seedlings and saplings by browsing soon after germination and, therefore, virtually absent; very abundant dung from grazing animals; bark stripped from young and mature trees and from branches on the ground; mosses scarce or absent; possible invasion of weed species, such as dock and sorrels (*Rumex* sp.) and meadow grasses (*Poa* spp.); the more palatable, grazing sensitive shrubs and herbs (e.g. *Lonicera periclymenum*, *Rubus fruticosus*, *Luzula sylvatica*, *Vaccinium myrtillus*) confined to inaccessible areas or, at least, noticeably more abundant there

Heavy – shrubs absent or moribund; ‘topiary’ effects on remaining shrubs; a browse line on mature trees; ground vegetation <20 cm tall with grasses, mosses or bracken (*P. aquilinum*) dominating; few patches of bare soil; surviving herb species usually dominated by unpalatable species, such as wood sorrel (*O. acetosella*) and bluebell (*H. non-scripta*); tree seedlings not projecting above ground vegetation height; abundant dung from grazing animals; bark stripping occasionally occurring; bulky, common mosses favoured at the expense of the rarer species requiring deeper shade and cover; the more palatable, grazing sensitive shrubs and herbs (e.g. *L. periclymenum*, *R. fruticosus*, *L. sylvatica*, *V. myrtillus*) confined to inaccessible areas or, at least, noticeably more abundant there

Moderate – patchy shrubs showing evidence of pruning or a browse line; ground vegetation variable in height up to 30 cm, comprising a mixture of grasses, herbs or dwarf-shrubs, including some of the more grazing-sensitive species (e.g. *L. periclymenum*, *R. fruticosus*, *L. sylvatica*, *V. myrtillus*) and showing direct evidence of browsing/grazing; localized close-cropped lawns where there is a concentration of grazing; patches of bare soil small and rare; tree saplings projecting above ground vegetation in a few areas; some dung from grazing animals; no bark stripping; wide range of moss species

Light – well-developed shrub layer, with no obvious browse line; a lush ground vegetation in places where the shrub layer covers not more than ca. 30–50% of the ground, dominated by grazing-sensitive species (e.g. *L. periclymenum*, *R. fruticosus*, *L. sylvatica*, *V. myrtillus*); tree saplings common in gaps; dung and tracks of grazing animals difficult to find; no bark stripping; browsed shoots scarce and localized, or totally absent; deep litter layer; ground mosses uncommon and consisting of few species

None – as for *Light* but with no browse line on the shrub layer; none, or very few, saplings where there has been no grazing for many years; no herbivore dung or tracks present; no browsed shoots; none, or very few seedlings; extensive mono-specific mats of vigorous ground-layer species, such as *Deschampsia flexuosa* may also occur on some sites

observable features of the ground-layer and shrub-layer components as well as on indicators associated with bark stripping and amounts of dung. The use of these indicators, thus, does not rely solely on assessing the amount and/or frequency of tree regeneration. Mitchell and Kirby (1990) conclude that, in general, grazing levels which produce a woodland with maximum nature conservation value will lie around the moderate level. Obviously this general rule will not always apply since some plant or animal species may do better in either an open, heavily-grazed, habitat or in a completely closed, ungrazed habitat. Reid (1996) concludes that the grazing and browsing pressure most suited to a particular woodland will be site specific and will be determined by the objectives for the site.

A range of indicators of habitat condition which relate to grazing and browsing pressure can thus be

used to help assess the current impact of ungulate herbivores on the nature conservation value of woodlands. However, since, in some cases, other factors may be responsible for the observed indicator level, an assessment which uses a large number of indicators is more reliable than one which uses one, or a few, indicators. The indicators given in Table 3 were developed for semi-natural woodlands in the British Isles and modifications would have to be made for those types of natural and semi-natural woodland found elsewhere. For example, beech (*Fagus sylvatica*) woodland will usually have very sparse ground-layer vegetation even under low, or no, grazing pressure. To gain an idea of trends in grazing and browsing pressure, the indicators can be categorized as being the result of long-term pressure (over years), medium-term pressure (over months), and short-term pressure (over days).

4.3. *How well does grazing pressure relate to nature conservation value?*

In general, a grazing pressure which is moderate in the long term, but may fluctuate between low and high in the short term, is likely to result in the maximum diversity of vertical structure within a woodland (see Table 3). Under these conditions the woodland will be able to perpetuate itself if there is enough disturbance to create regeneration niches for tree seedlings, but the browsing pressure is low enough to allow sufficient numbers of saplings to reach maturity. In a large area of woodland, spatial variation in grazing pressure around a mean moderate grazing level is also likely to increase horizontal structural diversity by allowing the full range of grazing-induced conditions to exist simultaneously. In small woodlands, it is unlikely that this sort of spatial variation in grazing pressures can exist. It would be expected that a woodland with the most diverse horizontal and vertical structure would offer the widest range of ecological niches and would, therefore, support the highest species diversity. Ferris-Kaan et al. (1998) describe a subjective, yet formalized, method of assessing the structural diversity of woodland and are currently working on testing the hypothesis that species diversity (including plant and animal species) is directly related to structural diversity. In many cases, managing for the highest species diversity will achieve the highest nature conservation value; however, several other factors have to be considered when assessing the latter. Accordingly, grazing levels prescribed for nature conservation purposes will not always deliver maximum species diversity.

4.4. *Measurement of damage caused by grazing*

The indicators listed in Tables 2 and 3, besides indicating grazing levels per se, can also be used to define whether optimum grazing levels are being achieved for nature conservation if one, or a range, of acceptable levels is determined. By definition, grazing levels outside the range will be causing damage. The measurement of damage thus involves choosing the most relevant, or easily assessed, indicators of those listed above and making an assessment in the field. This can be done quantitatively using a wide range of standard field-measurement techniques, e.g. Ratcliffe and Mayle (1992) for dung and Peterken

and Backmeroff (1989) for tree seedlings and saplings. Alternatively, where resources are limited, it can be done qualitatively by walking over the site and scoring it for a range of indicators. So far, such qualitative methods are in their infancy, although MacDonald et al. (1998) have produced a field guide for the identification of management impacts on upland habitats (excluding woodland but including scrub). They suggest methods by which such indicators might best be employed in the field for rapid habitat assessment. The Macaulay Land Use Research Institute and the Scottish Natural Heritage plan to test out a range of methods for applying the indicators in the field in Scotland, both at the patch and the site levels. This will be the first test of the repeatability of such methods and, therefore, of their accuracy and suitability as a monitoring technique.

5. Conclusions

Some concepts for objective and operational assessment systems which consider theoretical requirements have existed for some time. Improvements, in particular those relating to links as are required in modelling, need further research. On the one hand, there is a lack of basic knowledge, both concerning the manifold nature of the functions and possible impacts of the various ungulates on the forest ecosystem, and on the impact of forestry management practices and ‘habitat shaping’ on forest–ungulate interactions and ungulate damage to vegetation. On the other hand, there is considerable lack of clarity in specifying socio-economic and ‘cultural’ goals for the different forest areas if targets for each forest are to be meaningfully set. Without targets, neither ‘damage’ nor ‘benefit’ can be assessed. A particular difficulty lies in trying to forecast whether the final targets will be met in the distant future in terms of current ungulate-impact status.

In this context, it is noteworthy that statutory requirements are rarely given in operational terms and, thus, need to be interpreted. However, because of the differing policies and interests of the various interested parties (foresters, hunters, conservationists, the public, etc.), targets and threshold values set are based more on existing conventions and subjective viewpoints than on scientific understanding. In order to support a more scientific objectification of ungulate

impacts, key tasks for researchers are the development of an improved system of guidelines and the related methods of target setting in respect of stocking and regeneration, as well as of test criteria for the current status/target stock comparisons, aiming at a steady differentiation and refinement of threshold limits and appropriate surrogate (early warning) indicators.

Such studies should consider the forest ecosystem holistically, i.e. the forest floor with all its vegetation as well as all zoological aspects. In this context, long-term studies using fenced control areas in various forest systems having differing functions will be needed to support objective judgements regarding 'damage' and 'benefit' resulting from ungulate impact.

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